

System buying guide

Here is a list of systems with fairly decent overclocking headroom:

Budget System, around Rs 15,000

• AMD Athlon II X2 250	Rs 4,000
• Biostar A785GW – (ATI HD4200 Integrated Graphics).....	Rs 3,200
• Transcend 2GB DDR2 800 MHz RAM	Rs 1,200
• Western Digital 320 GB SATA2 Hard Disk	Rs 2,200
• Samsung/LG DVD Writer	Rs 1,200
• Gigabyte SuperB 460W SMPS.....	Rs 2,200
• Zebtronics Bijli	Rs 1,200
• OCZ Vanquishes AMD Edition (Optional).....	Rs 1,200
Total.....	Rs 16,400

Mid End System, around Rs 20,000-27,000

• AMD Phenom II X2 550 Black Edition	Rs 4,900
• Biostar T790GXBE	Rs 4,400
• 4GB Gskill Pi Black DDR2 800 MHz RAM	Rs 3,400
• Western Digital 1TB Green Hard Disk.....	Rs 4,200
• Samsung/LG/Lite On DVD Writer.....	Rs 1,200
• Corsair CX400 Seasonics 380W SMPS	Rs 3,200
• Antec 300 Cabinet.....	Rs 3,400
• Thermalright Ultima 90.....	Rs 1,900
• Scythe Sflex 65Cfm 120mm fan	Rs 500
Total.....	Rs 27,100

Mid End System Alternatives

• AMD Athlon II X4 620 Quad Core.....	Rs 5,500
• Coolermaster CM590 Cabinet.....	Rs 3,800
• ATI HD4850 NVIDIA GTS 250 (512MB) VGA adapter....	Rs 7,000

Mid-High End AMD System

• AMD Phenom II X4 955 Black Edition	Rs 11,000
• MSI GDFX-70	Rs 9,500
• OCZ AMD Black Edition Memory 4GB DDR3 1333 MHz RAM.....	~Rs 6,000
• Western Digital 1TB Black Edition Hard drive.....	Rs 5,400
• NVIDIA GTX 260	~Rs 10,000-11,000
• Samsung/LG/Lite On DVD Writer.....	Rs 1,200
• Corsair TX650 HX620 (modular) SMPS	~Rs 6,500 ~Rs 8,000
• Lancool PCK62 Cabinet	Rs 7,000
• Thermalright Ultra Extreme Black.....	Rs 3,400
• Scythe Sflex 75Cfm 120mm.....	Rs 500
Total.....	Rs 60,500

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heating up uselessly and reduces the electricity consumption. In the course of overclocking, we will be disabling CoQ if you have an AMD processor.

- **AMD C1E:** AMD C1E is a power management feature that suspends the system once it is idle.
- **Intel SpeedStep:** Intel SpeedStep is similar to AMD CnQ. Likewise, it reduces the system's power consumption by lowering the processor speed, when not in use.

Once you have disabled all the energy efficient techniques, it's time to start our business. Overclocking generally depends on the motherboard's capacity and capabilities; the better the motherboard, the better the processor will overclock. Of course, you need a capable processor, nonetheless.

To begin with, head on to the overclocking section of your BIOS (refer to the BIOS manual to locate this) and increase the FSB/HTT/BCLK by 2-5. If you have a special edition processor, then increase the multiplier by 1 from the default and save it. Return to Linux, and run the stress-testing program (we will learn about this later). If you have no errors or system performance issues, it means you have attained a stable overclock. Slowly and steadily, increase the value until you get to the point where the system stops responding. Kindly note, we have not yet increased the voltage of the system.

Once you have reached the limit, it's time to increase the voltage. A word of caution: Do not escalate the voltage value or you might end up with a fried CPU! Start by increasing from 0.05V and then continue increasing the BUS speed/multiplier till you get the maximum stable speed. Note: do not increase the voltage if you're using the factory shipped heat sink to cool the processor. Those cooling solutions are barely enough to cool the factory-shipped system, and increasing voltage on the default cooler might result in a dead processor. See the box titled 'After-market heat sinks' to get an idea